

# Package ‘GR2MSemiDistr’

August 4, 2025

**Type** Package

**Title** A package for hydrological modeling with a semi-distribute GR2M model adaptation in large-sample studies and operational systems.

**Version** 4.4

**Author** Harold Llauca <hllauca@senamhi.gob.pe>

**Maintainer** Harold Llauca <hllauca@senamhi.gob.pe>

**Description** This package run a semi-distributed GR2M model adaptation using the Weighted Flow Accumulation algorithm in TauDEM\_537 (required)

**License** GPL (>= 2)

**Encoding** UTF-8

**Depends** R (>= 4.0),

**Imports** airGR, foreach, hydroGOF, raster, terra, geos, rtop, tictoc, lubridate, sf, exactextractr, parallel,

**LazyData** true

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**NeedsCompilation** no

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|                       |                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create_Forcing_Inputs | <i>Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.</i> |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

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**Description**

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

**Usage**

```
Create_Forcing_Inputs(
  Subbasins,
  Precip,
  PotEvap,
  Qobs = NULL,
  DateIni,
  DateEnd,
  IniGrids = "01/1981",
  Save = FALSE,
  Update = FALSE,
  Members = NULL
)
```

**Arguments**

|           |                                                                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subbasins | A ‘SpatVector’ object containing the subbasins’ geometries. Must include the following attributes: ‘COMID’ (unique identifier), and ‘Region’ (string code for the hydro-climatic region). |
| Precip    | A ‘SpatRaster’ object with monthly precipitation data in [mm/month].                                                                                                                      |
| PotEvap   | A ‘SpatRaster’ object with monthly potential evapotranspiration data in [mm/month].                                                                                                       |
| Qobs      | Optional. A vector with observed streamflow in [m <sup>3</sup> /s] at the basin outlet.                                                                                                   |
| DateIni   | Start date of the output time series in ‘mm/yyyy’ format.                                                                                                                                 |
| DateEnd   | End date of the output time series in ‘mm/yyyy’ format.                                                                                                                                   |
| IniGrids  | Initial date of the gridded precipitation and evapotranspiration datasets in ‘mm/yyyy’ format.                                                                                            |
| Save      | Logical. If TRUE, saves the resulting data frame to a ‘.txt’ file in the ‘./Inputs’ directory. Default is FALSE.                                                                          |
| Update    | Logical. If TRUE, returns only the last month of data (useful for operational updates). Default is FALSE.                                                                                 |
| Members   | Optional. Integer indicating the number of ensemble members for streamflow forecasting. If provided, the date vector will be repeated accordingly. Default is NULL.                       |

**Value**

A data frame containing the model inputs in the airGR format, with columns:

**DatesR** Monthly dates

**P\_x** Mean precipitation for subbasin x, where x is the COMID

**E\_x** Mean potential evapotranspiration for subbasin x, where x is the COMID

**Q** Observed streamflow [m<sup>3</sup>/s], if provided

**References**

Aybar, C., Fernández, C., Huerta, A., Lavado, W., Vega, F., & Felipe-Obando, O. (2020). Construction of a high-resolution gridded rainfall dataset for Peru from 1981 to the present day. *\*Hydrological Sciences Journal\**, 65(5), 770–785. <https://doi.org/10.1080/02626667.2019.1649411> Llauca, H., Lavado-Casimiro, W., Montesinos, C., Santini, W., & Rau, P. (2021). PISCO\_HyM\_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *\*Water\**, 13(8), 1048. <https://doi.org/10.3390/w13081048>

**Examples**

```
library(GR2MSemiDistr)
data(pisco_pr) # Monthly precipitation SpatRaster
data(pisco_pe) # Monthly potential evapotranspiration SpatRaster
data(qobs)      # Data frame with observed streamflow
data(roi)       # Subbasins SpatVector

# Create model input data
data_inputs <- Create_Forcing_Inputs(
  Subbasins = roi,
  Precip = pisco_pr,
  PotEvap = pisco_pe,
  Qobs = qobs,
  DateIni = '01/1981',
  DateEnd = '12/2016',
  IniGrids = '01/1981'
)
View(data_inputs)
```

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Optim\_GR2MSemiDistr      *Model parameter optimization with the SCE-UA algorithm.*

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**Description**

Model parameter optimization with the SCE-UA algorithm.

**Usage**

```
Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  WarmUp = NULL,
```

```

Parameters,
Parameters.Min,
Parameters.Max,
Max.Functions = 5000,
Optimization = "NSE",
No.Optim = NULL
)

```

### Arguments

|                |                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data           | Dataframe with model input's data in airGR format from Create_Forcing_Inputs. (DatesR, P_1, P_2,...,P_n, E_1, E_2, ...E_n, Q). If Q is not available please provide only DatesR, P, and E.                                                                                                                                 |
| Subbasins      | Subbasins' shapefile. Must contain the following attributes: 'Area' (in km2), 'Region' (in letters), and 'COMID' (identifier number).                                                                                                                                                                                      |
| RunIni         | Initial date of the model simulation in 'mm/yyyy' format.                                                                                                                                                                                                                                                                  |
| RunEnd         | Ending date of the model simulation in 'mm/yyyy' format.                                                                                                                                                                                                                                                                   |
| WarmUp         | Number of months for the warm-up period. NULL as default.                                                                                                                                                                                                                                                                  |
| Parameters     | Vector of initial model parameters (X1 and X2) and correction factors of P (fp) and E (fpe) in the following order: c(X1, X2, fp, fpe). In the case of existing more than one 'Region' (e.g. regions A and B) please provide model parameters in the following order: c(X1_A, X1_B, X2_A, X2_B, Fp_a, Fp_B, Fpe_A, Fpe_B). |
| Parameters.Min | Vector of minimum values of GR2M model parameters and correction factors in the following order: c(X1_min, X2_min, fp_min, fpe_min).                                                                                                                                                                                       |
| Parameters.Max | Vector of maximum values of GR2M model parameters and correction factors in the following order: c(X1_max, X2_max, fp_max, fpe_max).                                                                                                                                                                                       |
| Max.Functions  | Maximum number of function evaluation for optimization. 5000 as default.                                                                                                                                                                                                                                                   |
| Optimization   | Objective function for optimization (NSE, KGE, or RMSE).                                                                                                                                                                                                                                                                   |
| No.Optim       | Regions not to be optimized. NULL as default.                                                                                                                                                                                                                                                                              |

### Value

List of optimal GR2M model parameters for each 'Region'.

Param: Best set of GR2M model parameters (sorted by 'Region').

Value: Final value of the objective function.

### References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. PISCO\_HyM\_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). Water. 2021; 13(8):1048. <https://doi.org/10.3390/w13081048>

### Examples

```

# Optimize GR2M model parameters for a single 'Region' using the KGE metric
optim <- Optim_GR2MSemiDistr(Data=data,
                             Subbasins=roi,
                             RunIni='01/1981',
                             RunEnd='12/2002',
                             WarmUp=36,

```

```

Parameters=c(1000, 1, 1, 1),
Parameters.Min=c(1, 0.01, 0.8, 0.8),
Parameters.Max=c(2000, 2, 1.2, 1.2),
Max.Functions=1000,
Optimization='KGE')

best_param <- optim$Param

```

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Routing\_GR2MSemiDistr *Routing discharges for each subbasin.*

---

## Description

Routing discharges for each subbasin.

## Usage

```

Routing_GR2MSemiDistr(
  Model,
  Subbasins,
  Dem,
  AcumIni = NULL,
  AcumEnd = NULL,
  Save = FALSE,
  Update = FALSE
)

```

## Arguments

|           |                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|
| Model     | List of model results from Run_GR2MSemiDistr.                                                                                         |
| Subbasins | Subbasins' shapefile. Must contain the following attributes: 'Area' (in km2), 'Region' (in letters), and 'COMID' (identifier number). |
| Dem       | Digital elevation model raster for the extent of the basin.                                                                           |
| AcumIni   | Initial date of the model routing in 'mm/yyyy' format. NULL as default                                                                |
| AcumEnd   | Ending date of the model routing in 'mm/yyyy' format. NULL as default                                                                 |
| Save      | Boolean to save results as a text file in the 'Outputs' location. FALSE as default.                                                   |
| Update    | Boolean for the updating mode where only the last month's values will be returned. FALSE as default.                                  |

## Value

List of model routing outputs.

QR: Routed discharge timeseries for all subbasins in [m3/s].

Dates: Vector of dates of the simulation period.

COMID: Vector of identifier numbers for each subbasin.

## Author(s)

Llaucha H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. PISCO\_HyM\_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*. 2021; 13(8):1048. <https://doi.org/10.3390/w13081048>

## Examples

```
# Load data
library(GR2MSemiDistr)
data(dem)

# Routing discharges in the streamflow network
rou <- Routing_GR2MSemiDistr(Model=model,
                             Subbasins=roi,
                             Dem=dem,
                             AcumIni='01/1981',
                             AcumEnd='12/2016')

View(rou$QR)
```

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|                   |                                                   |
|-------------------|---------------------------------------------------|
| Run_GR2MSemiDistr | <i>Run the GR2M model for multiple subbasins.</i> |
|-------------------|---------------------------------------------------|

---

## Description

This function runs the GR2M monthly water balance model on a semi-distributed basis across multiple subbasins, applying region-specific parameters and correction factors to precipitation and potential evapotranspiration.

## Usage

```
Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  WarmUp = NULL,
  IniState = NULL,
  Save = FALSE,
  Update = FALSE
)
```

## Arguments

|            |                                                                                                                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data       | Data frame of model inputs in airGR format created with <code>Create_Forcing_Inputs</code> . Columns should include <code>DatesR</code> , followed by <code>P_1</code> to <code>P_n</code> , <code>E_1</code> to <code>E_n</code> , and optionally <code>Q</code> . |
| Subbasins  | A 'SpatVector' object containing the subbasins' geometries. Must include the following attributes: 'COMID' (unique identifier), and 'Region' (string code for the hydro-climatic region).                                                                           |
| RunIni     | Initial simulation date in 'mm/yyyy' format.                                                                                                                                                                                                                        |
| RunEnd     | Final simulation date in 'mm/yyyy' format.                                                                                                                                                                                                                          |
| Parameters | Numeric vector of model parameters and correction factors. Format: <code>c(X1_region1, ..., X1_regionN, X2_region1, ..., Fpp_regionN, Fpe_region1, ..., Fpe_regionN)</code>                                                                                         |
| WarmUp     | Optional number of months to discard as warm-up. Default is <code>NULL</code> .                                                                                                                                                                                     |
| IniState   | Optional list of initial state values for each subbasin. Default is <code>NULL</code> .                                                                                                                                                                             |

|        |                                                                                             |
|--------|---------------------------------------------------------------------------------------------|
| Save   | Logical. If TRUE, outputs will be saved as text files in the 'Outputs/' folder.             |
| Update | Logical. If TRUE and Save = TRUE, overwrites previous month's files before saving new ones. |

### Value

A list with the following elements:

|          |                                                                                       |
|----------|---------------------------------------------------------------------------------------|
| PR       | Matrix of precipitation [mm/month] per subbasin                                       |
| AE       | Matrix of actual evapotranspiration [mm/month] per subbasin                           |
| SM       | Matrix of soil moisture [mm/month] per subbasin                                       |
| RU       | Matrix of runoff [mm/month] per subbasin                                              |
| QS       | Matrix of simulated discharge [m3/s] per subbasin (not routed)                        |
| Dates    | Vector of dates used in the simulation                                                |
| COMID    | Vector of subbasin COMID identifiers                                                  |
| EndState | List of final model states for each subbasin                                          |
| SINK     | Optional. Data frame of simulated and observed streamflow at outlet, if Q is provided |

### References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO\_HyM\_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

### Examples

```
# Load input forcing data and subbasins (see Create_Forcing_Inputs)
model <- Run_GR2MSemiDistr(
  Data = data,
  Subbasins = roi,
  RunIni = '01/1981',
  RunEnd = '12/2016',
  Parameters = c(10.976, 0.665, 1.186, 1.169)
)

# View model outputs
View(model$PR) # Precipitation [mm/month]
View(model$QS) # Simulated discharge [m3/s] per subbasin (not routed).
print(model$SINK$SIM) # Simulated outlet discharge
print(model$SINK$OBS) # Observed outlet discharge (if available)
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